

Design Documentation: YouTube Studio Agent

1. Purpose of the Application

YouTube Studio Agent is a private, internal dashboard developed exclusively for my personal use as an independent YouTube creator. It will not be distributed or sold to any third parties.

The primary goal of this tool is to centralize my YouTube channel management. It integrates YouTube Data API and Google Ads API to provide a unified dashboard where I can monitor organic metrics alongside paid ad campaign performance.

2. How the Google Ads API will be used

The application will utilize the Google Ads API strictly for reading reporting metrics and toggling campaign statuses for my own video campaigns. Specifically, the API will be used to:

- Fetch active and paused video promotion campaigns.
- Retrieve performance metrics such as Impressions, Clicks, and Cost per campaign.
- Allow me to pause or activate a campaign directly from the dashboard.

3. User Interface Mockup

Below is a visual representation of how the Google Ads data will be displayed within the internal dashboard. The user securely logs in with their own Google credentials (OAuth 2.0).

Promotions Dashboard

Launch Promotion

Video Thumbnail	My Latest Vlog Episode	ACTIVE
	Budget: \$10.00/day	
	Impressions: 1,250	Clicks: 85
		Cost: \$5.20

Video Thumbnail	Tutorial: How to Code	PAUSED
	Budget: \$15.00/day	
	Impressions: 4,500	Clicks: 310
		Cost: \$18.40

4. Technical Implementation & Security

The dashboard is built using Nuxt.js. It operates on a local environment or a private VPS. It does not store user data publicly. Authentication is handled via standard OAuth2 protocols directly through Google's authorization servers, ensuring that the Developer Token and access credentials remain completely secure.